No.

Date

Karakteristik Fiter

Jurnal High Frequency

Kelompok : - Bimo Raul S (211344009)
- Ditto Nur A (211344010)
- Dwi Agustina (211344011)
- Dyelvio Wijoyo (211344012)

* Low Pass Filter

o> Setup

Lampiran Gambar 1.1.

o> Gambar Respon Filter

Lampiran Gambar 1.2

o> Perhitungan

Ripple Max : 1.125 MHz ; -3,8 dBm

Ripple Min : 1.050 MHz ; -8,9 dBm

Frequency Cutoff : 1.170 MHz

① Ripple = Ripple Max - Ripple Min

= -3,8 - (-8,9) = 5,1 dBm

② Insertion Loss = Ref - Ripple Tengah

= 0 - (-6,8) = 6,35 dBm

③ Shape Faktor = $\frac{\text{Bw redaman besar}}{\text{Bw redaman kecil}}$

= $\frac{1.525 \text{ MHz}}{1.170 \text{ MHz}}$ = 1,303

④ Ripple Tengah / Flatness = $\frac{\text{Ripple}}{2}$ + Ripple Min

= $\frac{5,1}{2}$ + (-8,9)

= -6,3 dBm

⑤ Frequency Cutoff = 1.170 MHz

* Band Pass Filter

o> Setup Pengukuran

Lampiran Gambar 2.1

o> Gambar Respon Filter

Lampiran Gambar 2.2

o> Perhitungan

fc atas = 1 901 MHz

fc bawah = 1.887 MHz

No.

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① Insertion Loss = REF - Flatness

= 0 - (-9,4 dBm)

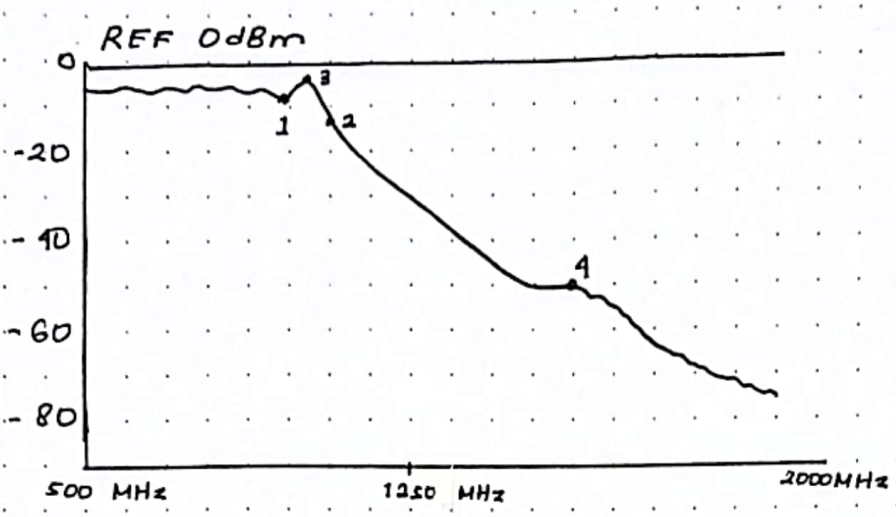
= 9,4 dBm

② Shape Faktor = $\frac{\text{BW redaman besar}}{\text{BW redaman kecil}}$

= $\frac{1914,5 - 1874}{1901 - 1887}$

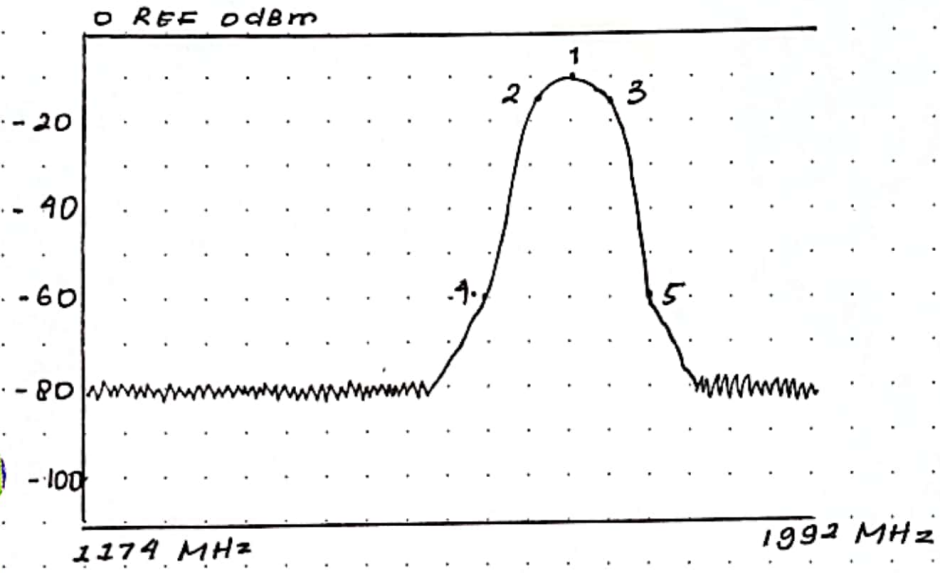
= $\frac{40}{14}$ = 2,86

※ Gambar 1.2.



o> M1 = 1050 MHz ; -8,9 dBm
o> M2 = 1170 MHz ; -12,1 dBm
o> M3 = 1125 MHz ; -3,8 dBm
o> M4 = 1525 MHz ; -48,6 dBm

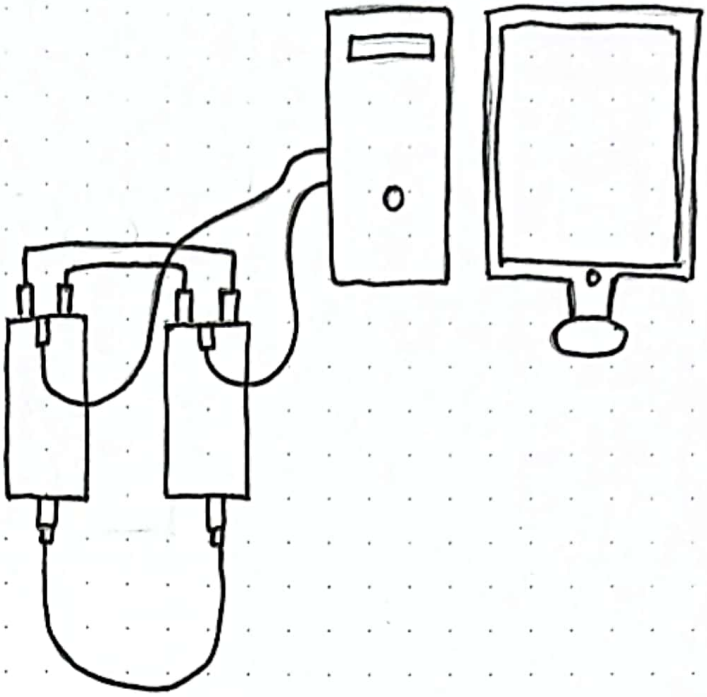
※ Gambar 2.2

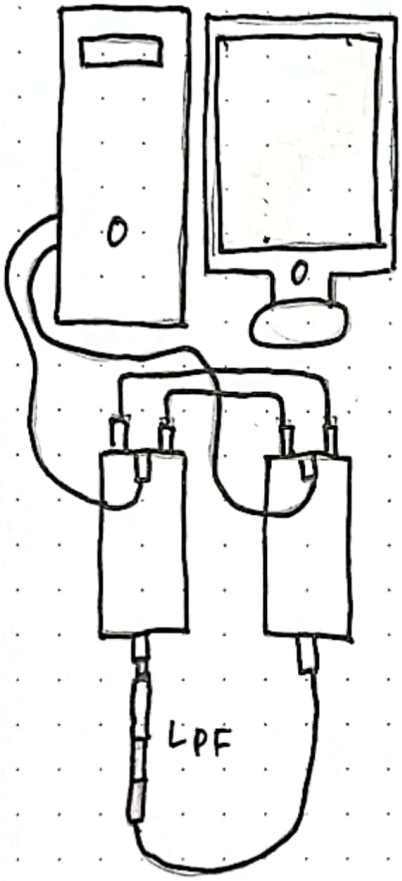


o> M1 = 1893,5 MHz ; -9,4 dBm
o> M2 = 1887,5 MHz ; -12,9 dBm
o> M3 = 1902 MHz ; -12,8 dBm
o> M4 = 1874 MHz ; -60 dBm
o> M5 = 1914,5 MHz ; -60,6 dBm

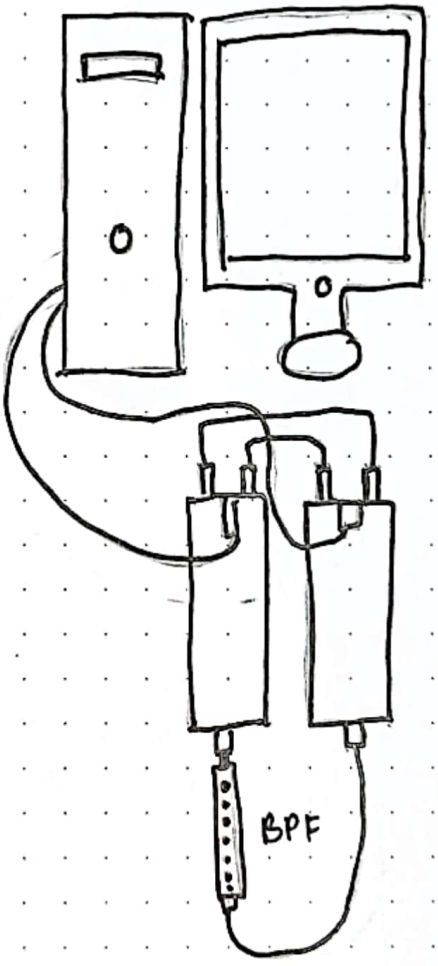
JOYKO

SETUP PENGUKURAN





GAMBAR 1.1



GAMBAR 2.1